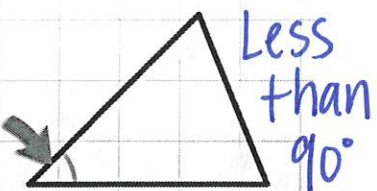
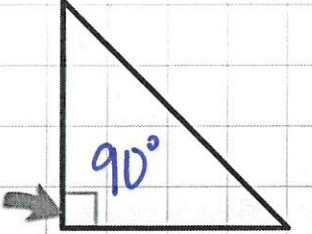
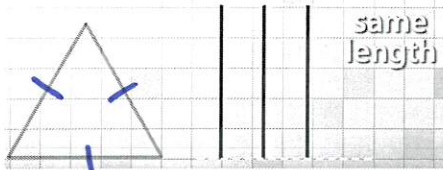
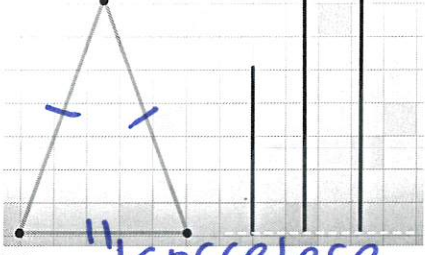
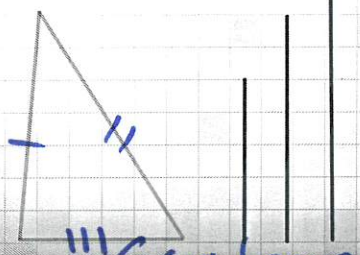
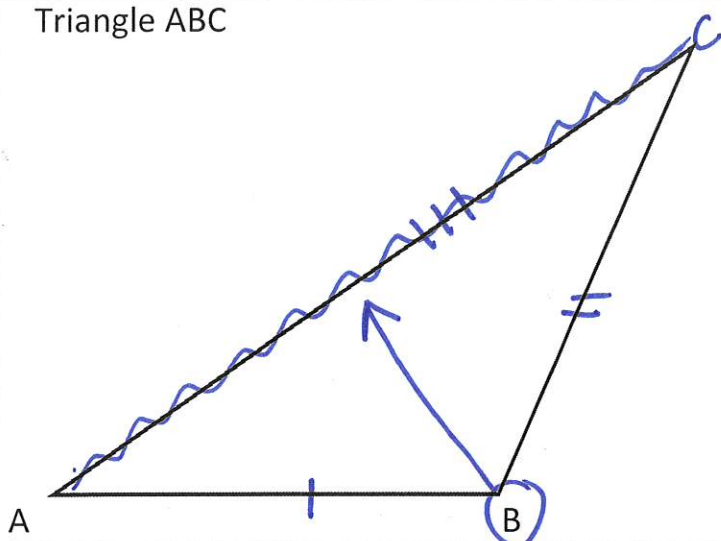


Unit 8 Triangles - Day 1: Triangles, Triangles, Triangles

Classify by Angles	 Name: <u>Acute</u>	 Name: <u>Right</u>	 Name: <u>obtuse</u>
	 Name: <u>Equilateral</u> <u>3</u> equal sides	 Name: <u>Isosceles</u> <u>2</u> equal sides	 Name: <u>Scalene</u> <u>NO</u> equal sides

Exploration:

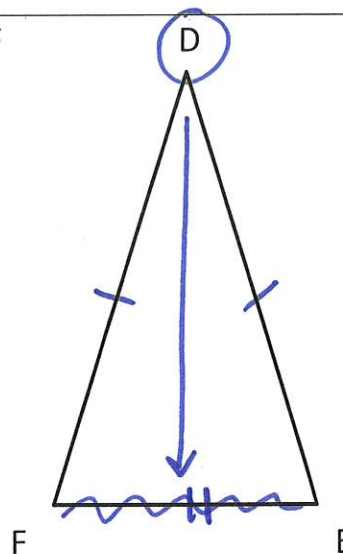
Triangle ABC



Triangle ABC ($\triangle ABC$)

Angle Measure	Side Length
$m\angle CAB = 35^\circ$	$\overline{BC} = 6.5 \text{ cm}$
$m\angle ABC = 114^\circ$	$\overline{AC} = 10.5 \text{ cm}$
$m\angle ACB = 31^\circ$	$\overline{AB} = 6 \text{ cm}$

Triangle DEF



Triangle DEF ($\triangle DEF$)

Angle Measure	Side Length
$m\angle FDE = 34^\circ$	$\overline{EF} = 3.5 \text{ cm}$
$m\angle DEF = 73^\circ$	$\overline{DF} = 6 \text{ cm}$
$m\angle DFE = 73^\circ$	$\overline{DE} = 6 \text{ cm}$

Based on the measurements above, this triangle can be classified as

obtuse scalene

Based on the measurements above, this triangle can be classified as

Acute Isosceles

What do you notice about the largest angle compared to the longest side?

The longest side is OPPOSITE the largest angle.

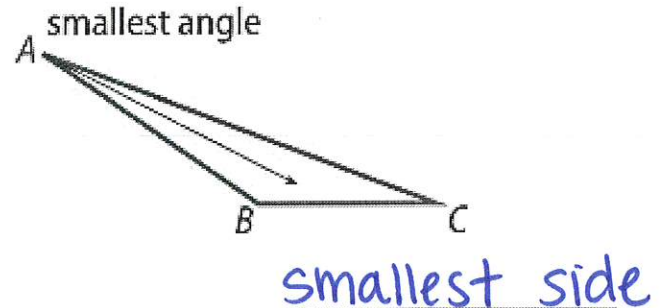
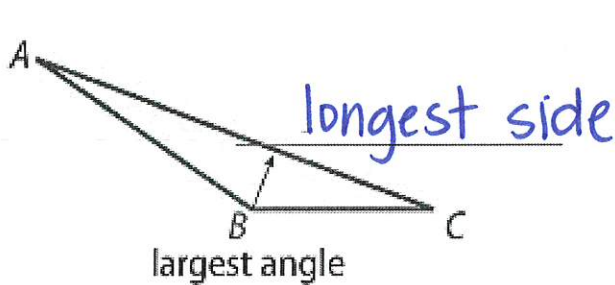
If a triangle has two congruent angles, what can you determine about the side lengths opposite those angles?

The sides are also congruent

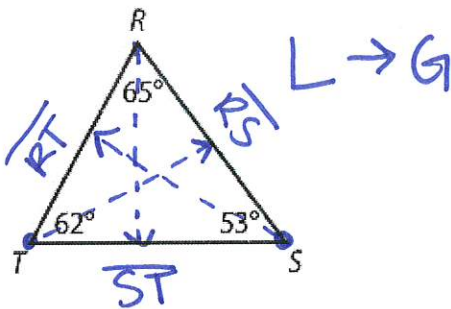
What do you notice about the position of the smallest angle compared to the shortest side?

The shortest side is OPPOSITE the smallest angle.

Order Sides and Angles of Triangles

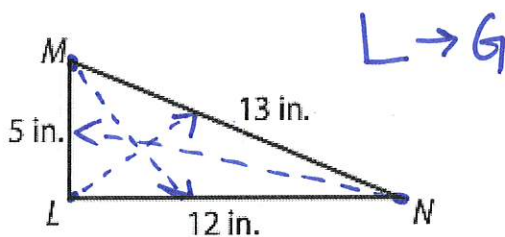


Order the sides and angles of the triangle from least to greatest side length:

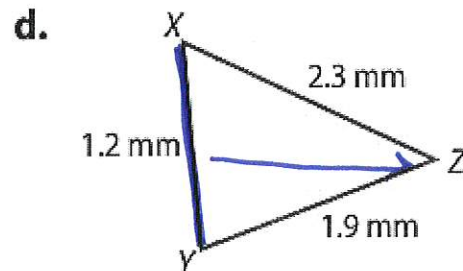
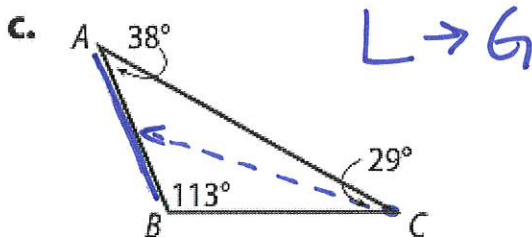


Angles: $\angle S, \angle T, \angle R$
Sides: $\overline{RT}, \overline{RS}, \overline{ST}$

Order the sides and angles of the triangle from least to greatest angle:



Angles: $\angle N, \angle M, \angle L$
Sides: $\overline{LM}, \overline{LN}, \overline{MN}$



Angles: C, A, B
Sides: $\overline{AB}, \overline{BC}, \overline{AC}$

Angles: Z, X, Y
Sides: $\overline{XY}, \overline{YZ}, \overline{XZ}$